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Evaluating the Non-linear Impact of Digital Literacy on Financial Inclusion: Evidence from Emerging Economies via Explainable Deep Learning

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Abstract
The link between digital literacy and financial inclusion is often oversimplified in existing studies that rely on linear frameworks. Such approaches frequently overlook the structural differences and non-linear patterns across various developing regions. In this study, we develop FERENet (Feature Ranking and Explainable Network), a hybrid model combining deep neural networks with random forests to analyze a dataset spanning 56 economies in Africa, Asia, and Latin America. Our results indicate that FERENet significantly outperforms traditional General Method of Moments (GMM) estimations, explaining 94% of the variance. By employing SHAP-based interpretability, we identify specific regional digital barriers that challenge the effectiveness of universal policy designs. The findings suggest that financial inclusion efforts should move away from generic literacy programs toward infrastructure-specific interventions that align with local economic conditions. This study provides a more nuanced understanding of how digital skills translate into financial access in constrained environments.

Keywords Deep Learning · Digital Inclusion · Digital Literacy · Emerging Economies · Financial Inclusion · Machine Learning · Regional Analysis

1 Introduction

Digital literacy, with the ability to locate, evaluate, and create information using digital technologies has become a fundamental skill for participation in today's knowledge-based economy [1]. Yet, despite rapid advances in connectivity, only about 40% f youth and adults worldwide achieve the minimum proficiency levels in digital literacy needed to fully engage with digital platforms and services [2]. Moreover, an estimated 2.6 billion people remain entirely offline, unable to access the benefits of digital inclusion from e-health and e-government to online education and financial services impairing global inequality and poverty [3]. Closing this divide is therefore a priority for sustainable development, as digital skills directly influence employment opportunities, economic resilience, and social well-being [4].

Artificial intelligence (AI) is reshaping digital literacy itself, offering powerful tools for learning, assessment, and personalized support [5]. In education, 92% f university students now report using AI in some form up



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sharply from 66% ust a year earlier with applications ranging from concept explanation to research idea generation [6]. Yet while AI tools proliferate, only about 32.3% of espondents demonstrate an adequate understanding of how AI decision-making processes work, highlighting a critical "AI literacy" gap even among educated users [7]. Equipping learners and communities with both digital and AI literacy is therefore essential to harness these technologies for equitable development, rather than deepening existing divides [8]. Figure 1 presents the digital divide between AI and Literacy gap.

The rationale behind this study is the necessity to find strong, evidence-based approaches to measure the digital literacy educational programs and measure their financial effects in the settings of rural communities that are frequently neglected in the context of AI and education studies [9]. Although the relationship between digital skills and income, as well as well-being, has been studied in the past, few studies have combined hybrid deep learning models with feature-ranking approaches [10] to establish the strongest indicators of economic performance among the elements of digital literacy. Also, specificity of targeting interventions is lacking rural populations are characterized by distinctive barriers, including infrastructure shortage or contextualized learning requirements [11].

The available methods of analysis have a few significant flaws. Classical econometric approaches, including the Generalized Method of Moments, are usually based on linear or weakly non-linear assumptions and are generally incapable of addressing the high-dimensional interaction between digital literacy, socio-economic variables, and financial access [12]. Such models also find it difficult to cope with heterogeneous patterns across countries where the structural differences among regions imply different relationships that cannot be satisfactorily modeled using fixed parametric forms. In recent machine learning systems, such as deep neural networks, CNN-LSTM hybrids, and tree-based learning systems such as XGBoost have tried to overcome some of these problems by modeling nonlinear relationships [13]. Nonetheless, most of these models are independent architectures, which do not have built-in support of robust feature selection, noise reduction, and interpretability. Specifically, they do not usually give explicit information about the importance of features, nor do they trade-off predictive performance with explainability, which is essential in policy-oriented research [14]. To fill these gaps, this study introduces FERENet (Feature Ranking and Explainable Network), a combination of ensemble-based feature refinement and residual deep learning. With feature selection based on the combination of the Random Forest and LightGBM, and a residual deep neural network with Huber loss, the proposed model can effectively represent nonlinear interactions and be robust to noisy socio-economic data. Moreover, SHAP-based interpretability

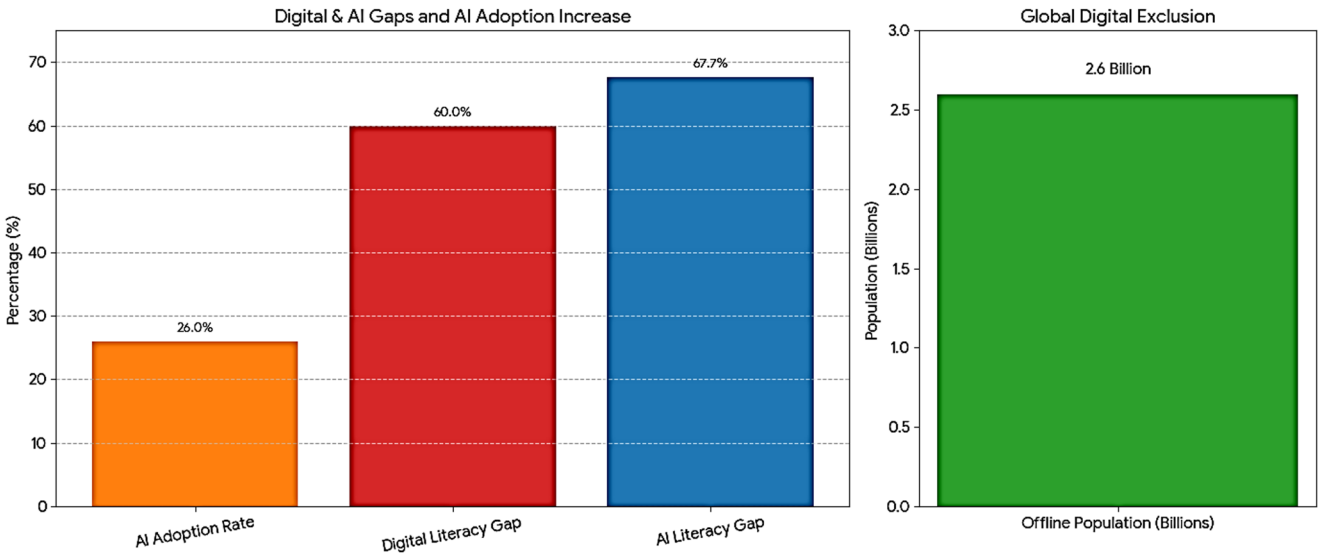


Fig. 1 Digital and AI gaps and global digital exclusion (compiled by authors based on data from ITU, World Bank, UNESCO, and global AI reports such as McKinsey and Stanford AI Index)

has been integrated such that it can clearly identify key financial inclusion drivers across regions. The primary contribution of the work is in the possibility to bridge the gap between predictive accuracy and interpretability in cross-country financial inclusion modeling and explicitly consider regional heterogeneity and the interplay of complex features that are frequently ignored in traditional and standalone machine learning methods.

In contrast to traditional ensemble pipelines, FERENet proposes a structured hierarchical learning process with joint optimization of feature transformation, selection, and prediction using a staged structure. The uniqueness of the details is not in the single elements, but in the chain of their combination and arrangement. In particular, the dimensionality is controlled and the model exhibits non-linear socio-economic interactions by using the following features: (a) the expansion of features using polynomials and (b) the mutual selection of information. Moreover, the addition of dual ensemble refinement layer (Random Forest and LightGBM) before residual deep learning enables the mining of hierarchical feature importance which is utilized by residual DNN to acquire higher-order dependencies. This stacked learning methodology coupled with Huber loss optimization offers robustness to noisy real-world data- a feature that is not considered in standard econometric and machine learning models. The key contributions of the study are given in the following.

- Developed a residual learning-based framework integrating ensemble methods with polynomial feature expansion and mutual information selection to capture complex interactions and reduce noise.
- Employed a Huber loss-based optimization strategy within a residual DNN to improve robustness against outliers in socio-economic survey data.
- Designed a hybrid ensemble layer (Random Forest + LightGBM) as part of the feature refinement process before final deep learning prediction.
- Demonstrated how interpretable feature importance and residual deep learning can jointly support evidence-based policy analysis using national datasets.

The rest of this study is structured in the following manner. Literature Review examines the theoretical background of both digital and AI literacy and the socioeconomic consequences. Materials and Methods section outlines the hybrid deep learning and feature-ranking model that was created in this research, model framework and data sources. Experimental Setup section provides details of the data collection and pre-processing to be used in the analysis of rural education programs and economic indicators. Results and Experimental Findings section has experimental results and examines the performance of the models; the main predictive features and economic impacts estimation and Conclusion section provides a final part of the study (conclusion) and recommendations of future research and practice.

1.1 Literature Review

Recent research consistently indicates that digital literacy is an essential tool to develop rural areas. By national survey data collected in China, Wang et al. [11] found that greater digital literacy can greatly reduce income gap in rural farmers due to much more non-agricultural work and labor migration. This mirrors the "normalization theory" of the digital divide, which posits that digital literacy helps lower-income farmers advance economically. Similarly, Zhou et al. [15] Digital literacy among former poor households helps prevent re-falling into poverty, largely by opening opportunities for local off-farm and migrant jobs. Zhang et al. [16] found that an increase of one standard deviation in digital skills increased rural household income by 15.7% and reduced relative poverty with 12.3%. The impact was particularly felt in Western China and in vulnerable communities, demonstrating that increasing rural digital literacy can be a force for inclusive growth.

However, the literature also reveals a significant tension regarding how these gains are distributed. Liu and Zhou [17] found that although digital literacy increased village average income in China, it increased intra-rural income inequality, disproportionately benefiting those with higher incomes. They also concluded that digital literacy was more beneficial for the growth in wages and agribusiness income of the rich farmers, and so affecting rural intra-income disparity negatively. This divergence in findings suggests that the impact of digital education

is not universally positive but is highly context specific. This is different from Wang et al. [18] narrower gaps found that the equity impact of digital literacy might be context-specific and contingent on other policies. The two studies both agree, however, that lacking digital skills continues to prevent many rural residents from accessing new sources of income (e.g. e-commerce, online services), highlighting the importance of broad-based digital inclusion initiatives. Indeed, He et al. [19] highlight that the government should focus on improving rural network endowment and digital education level, to let the disadvantage group to share the "Digital dividend" rather than be left behind. Lechman & Popowska et al. [20] further assert that use of digital technologies is an imperative for poverty alleviation in these LDCs, yet it will necessitate closing the access and capability divides. Taken as a whole, these works show that digital literacy can be a double-edged sword – an engine of rural prosperity if widely acquired, or a generator of new inequalities if enjoyed only by elites.

Beyond theoretical impacts, empirical interventions have demonstrated the transformative power of targeted digital training. Efforts to increase digital literacy for rural people have had some success. In India, study conducted by Gogoi et al. [21] which is quasi experimental in nature studied a three-year digital literacy project for adolescent girls in far-flung villages. They discovered that the portion of girls with above-median digital literacy scores rose from about 41% before the intervention to 90% at post-program, a dramatic increase compared to a control group. Villages that received treatment had higher continuation of girls' education, greater access to ICT devices, increased knowledge of e-gov't services, and higher perception of personal security. COVID-19 highlighted that digital preparedness is a critical component of community resilience. The value of such preparedness became even more evident during global disruptions. Marsdenia et al. [22] recorded how the moderate digital literacy of rural tourism entrepreneurs in Indonesia enabled them to swiftly transition to online business during lockdown and keep the local economy running. Communities that had at least basic digital skills were better able to access information, market products from a distance and cope with disruptions. Fahmi and Savira [23] did case studies in Indonesian villages and showed that, even in contexts where digital uptake was slow to develop, individuals who knew how to use online tools had a disproportionate impact on collective coping mechanisms. Despite these successes, the transition to a digital economy faces systemic hurdles. Choudhary et al. [24]evaluated the country's national Digital Literacy Training Programs (DLTPs) in rural villages based on interviews with management personnel, trainers and trainees. Key identified problems are uneven access to electricity and internet; lack of devices; low digital literacy; and socio-cultural biases (e.g. older peoples, women) being left behind. There was a power dynamic — the more educated local representatives frequently conceived of the programs without fully understanding the needs of the marginalized groups. These results are consistent with Maulana et al. [25] highlighting that high-level technology introduction alone will not do – systemic support and locally situated approach needed to be established to empower rural learners as well. When Deep Learning tools and digital resources are integrated mindfully into education, they can help reduce the barriers to accessing information, encourage critical thinking, and "create a generation of learners who are adaptive, reflective, ethical".

A new trend in applications of ML and hybrid deep learning models to assess digital literacy and its effects can be observed. As the limitations of traditional support systems become clear, a new trend has emerged: using Machine Learning (ML) and hybrid deep learning to assess literacy and its effects. These data-based methods can reveal patterns and have predictions that are beyond what traditional analyses can achieve. For example, Park and Feng [26] proposed a classification model that estimates older adults likely to be excluded from the digital cohort in South Korea. Their model, trained on a national survey dataset, pinpointed the most important influences (demographics, internet usage, self-efficacy, social connectedness) on whether an older adult is digitally divided. Benzer et al. [27] showed that ML can be used to automatize the assessment of teachers' digital competences. They gathered 172 science teachers' survey data on digital literacy in Turkey and trained two classifiers—an SVM (with Sequential Minimal Optimization) and a multilayer perceptron neural network—to classify teachers into different literacy levels. The SVM had the best accuracy (approx. 92%), better than the neural network in profiling the proficiency level. It also showed that the advanced digital literacy skills were possessed by a small group of teachers, everyone else had only average digital literacy, reinforcing the necessity for age targeted upskilling. This shift toward algorithmic assessment has also highlighted the importance of identifying

which specific factors drive literacy. Another study of the elderly population was conducted by Kong and Wang [28] who used ensemble learning algorithms (Random Forest, CatBoost, XGBoost etc.) to predict their digital literacy levels. The best accuracy of ~ 86.2% was achieved with CatBoost, just a bit above the other boosting techniques. Most importantly, and through the application of feature importance and SHAP values, they uncovered predictors that play a dominant role on the final predictive model – beyond age and education, having social support and a community digital environment affects considerably the digital literacy of elderly. Malik et al. [29] proposed a new ensemble feature selection model DE-FS ("Dynamic Feature Ensemble-Evolution"), to predict students' performances in different educational databases. DE-FS is a hybridized approach combining traditional filters (correlation analysis, information gain and chi-square) with a dynamic thresholding scheme which adjusts according to the input data changing pattern. Their feeding the most informative features to classification models obtained superior accuracy and robustness than baseline methods in various student performance tasks. The authors observe how the adaptive thresholds to features of DE-FS helped to reduce overfitting and noise for better predictive precision, hence showing the strength of hybridizing expert feature selection with the flexibility of deep learning. The intersection of digital literacy and financial inclusion is now being explored through increasingly complex architectures. In the healthcare sector, Huang et al. [30] extracted disease names from the electronic health records of chronic hepatic diseases by means of pattern matching. A two-stage hybrid deep learning approach proposed to predict the negative behavioral consequences of childhood cancer in young cancer survivors. They considered a multimeric voting filter to order medical, socio-environmental, and treatment attributes and embedded a deep neural network with heavy dropout as a selector for the top attributes for each outcome. Yang et al. [31] using an instrumental-variable Tobit estimation model measured how it leads to the prevention of rural household 'Internet information poverty. Shao et al. [32] protective effect was particularly evident for younger and less educated farmers, implying that digital literacy can in some sense serve as a replacement for formal education in reducing poverty risk. The base study by Nabil Adel [33] analyzes how digital literacy and blockchain adoption affect financial inclusion across 56 emerging countries in Africa, Asia, and Latin America from 2011 to 2021 using a two-step system GMM model. It finds that digital literacy boosts financial inclusion in Africa but surprisingly hinders it in Asia, while new technology adoption consistently promotes inclusion in Asia and Latin America. These methods enable researchers to forecast results, know how things cause one another to happen, and to customize intervention to a greater degree than ever imagined in the past. The practical implications of such data-driven insights for scaling digital literacy programs are highly relevant for realizing their societal and economic potential.

Recent studies have increasingly explored the intersection of digital literacy, financial inclusion, and advanced machine learning techniques. Specifically, in the realm of financial inclusion, literature is moving toward a synthesis of predictive power and interpretability [34]. For instance, explainable and hybrid machine learning approaches have gained prominence, with I. Choi et al. [35] proposing a network-based exploratory data analysis combined with deep clustering for financial profiling, emphasizing the importance of interpretable multi-stage learning. Similarly, X. Kong and Wang [36] demonstrate the effectiveness of ensemble learning integrated with SHAP-based feature importance for analyzing digital literacy using survey data, reinforcing the value of explainability in socio-economic modeling. In the context of behavioral prediction, Ahmed et al. [37] employ explainable machine learning to model mobile payment usage, linking application behavior with financial engagement. Earlier work by D. Mhlanga [38] highlights the growing role of artificial intelligence in financial inclusion, particularly in credit risk assessment within emerging economies, while P. K. Ozili [39] provides a critical discussion on the structural and methodological challenges in establishing causal relationships in financial inclusion research. Additionally, another study [40] focus on predicting digital twin cities exclusion using national survey data, underscoring the importance of demographic and access-related factors. Collectively, these studies motivate the need for hybrid, explainable, and region-aware modeling frameworks, which the proposed FERENet approach aims to address by integrating ensemble feature learning with residual deep neural networks for improved predictive performance and interpretability.

It is important to emphasize that the comparison between FERENet and the GMM estimator should not be interpreted as a direct methodological superiority test. The GMM, as used in is designed for causal inference under structural econometric assumptions and aims to identify consistent parameter estimates rather than maximize predictive accuracy. In contrast, FERENet is a supervised learning framework optimized for predictive performance and generalization. Therefore, differences in R^2 values reflect differences in modeling objectives rather than strict methodological superiority.

2 Materials and Methods

The Proposed Methodology section describes the FERENet framework. It combines deep neural networks, random forests, and explainability techniques. The FERENet framework is categorized as a Hybrid-Ensemble Deep Learning Architecture. It is "Hybrid" in its integration of traditional statistical feature selection with neural computing, and "Ensemble" in its use of multiple gradient-based and tree-based learners [41]. The section also explains the algorithm used to optimize the model's performance. The design of the FERENet framework follows recent advancements in high-dimensional predictive modeling, which emphasize the integration of ensemble architectures with rigorous explainability to evaluate complex organizational or socio-economic efficiencies [42].

2.1 Proposed Framework

The framework shown in Fig. 2 is the FERENet model that is applied to assess digital literacy and its economic effects in rural populations. The presented FERENet architecture incorporates ensemble learning, residual deep neural networks, and feature interpretability to assess the economic value of digital literacy. The workflow, as shown in Fig. 2, starts with the data preparation, where raw survey data of the Global Findex Database 2021 are loaded, cleaned, and preprocessed. Target variables are selected, null rows are removed, numerical features are standardized, and categorical variables such as Region and Income Group are label encoded. This is followed using a nonlinear interaction between predictors by expanding the form of the feature polynomials and then selecting the most informative features using mutual information scores [43]. The chosen attributes are then fed into a hybrid ensemble model consisting of Random Forest, Residual DNN, and LightGBM. The Random Forest models nonlinear correlations and offers initial feature relevance, the Residual DNN models the intricate interactions and reduces overfitting with the aid of residual connections and dropout, and LightGBM optimizes the predictions with the help of gradient boosting, supplementing the other elements. Lastly, the importance of features is recorded and read out through permutation importance and SHAP-based analysis and LIME is applied on the selected cases to give interpretable explanations on the most important features [44]. This integrated architecture guarantees strong predictive performance, transparency and the determination of key drivers in regions, which is evidence-based policy. To make the methodology clear, the training processes of various model elements within the proposed FERENet framework are clearly separated. Tree-based ensemble models such as Random Forest and LightGBM are trained on the whole training set without an epoch-based optimization. Such models are based on ensemble learning using decision trees and boosting mechanisms instead of gradient-based updates at each epoch. Contrastingly, the deep neural network (DNN) [45] component is an iterative optimization with mini-batch gradient descent. The DNN is learned in repeated epochs, each epoch is a full cycle over the training set. The Adam optimizer is used to minimize the Huber loss function to be robust to outliers.

More specifically to explain the flow of model a real-world data is subjected to the model. It is a modular model that is based on four key phases: data preparation, feature engineering, ensemble model training, and feature interpretation. The modules are modeled to tackle the structure and issues of the dataset. Moreover, it is in line with the aims of the research of education and economic dimensions of digital literacy. Data is loaded into the system in the data preparation phase. Once the data is cleaned, the next step is the feature engineering step. The reason behind this step is to add more meaningful features onto the dataset. Expansion of polynomials of

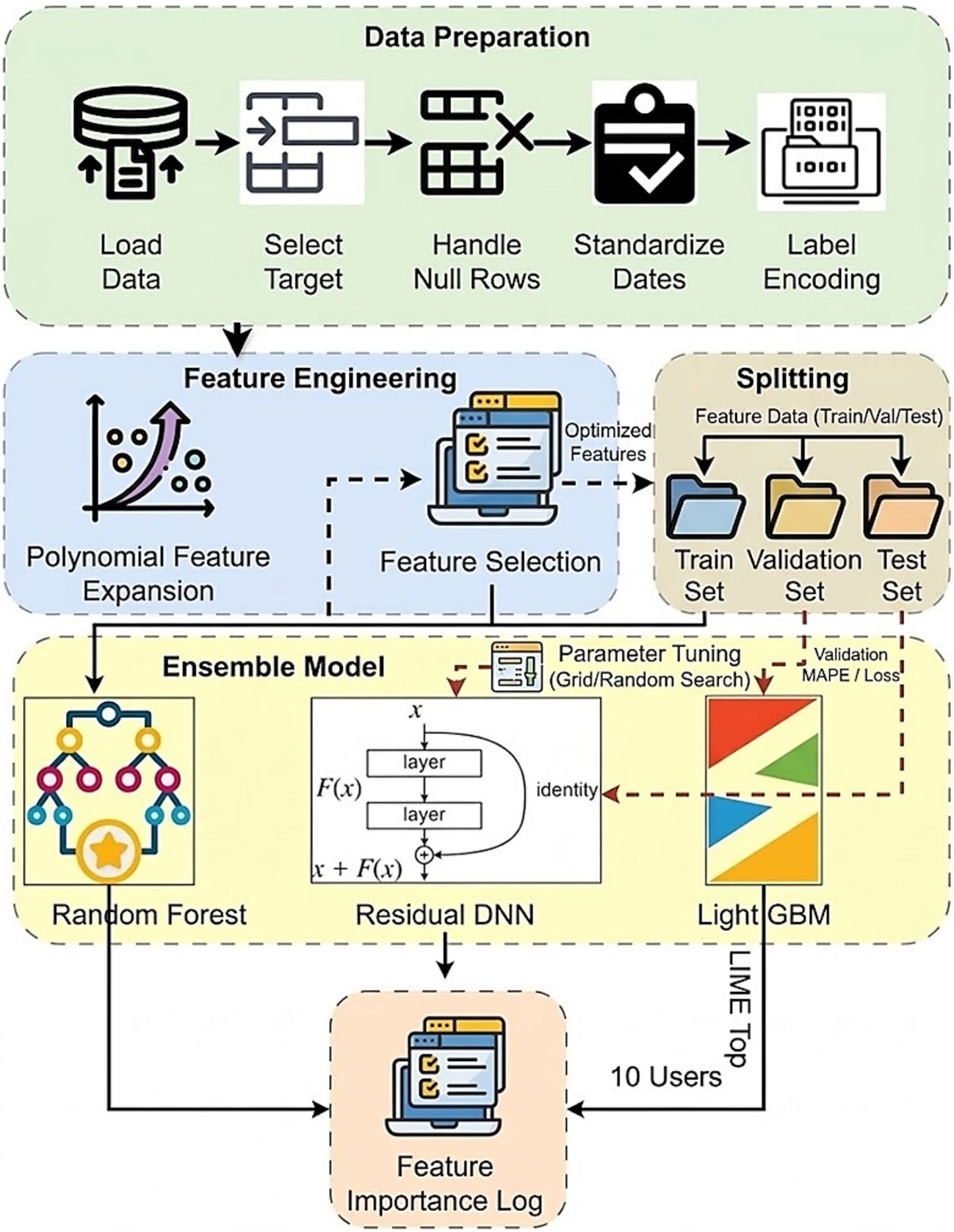


Fig. 2 FERENet Framework for Evaluating Digital Literacy and Economic Impact

features is done to express non-linear interaction between the variables. The step is significant as digital literacy is frequently based on a set of variables, including age and education or gender and employment. A selection mechanism is then applied after feature set is expanded. The feature selection can be used to select the most predictive features and minimize noise.

The ready data is then sent to ensemble modelling stage. FERENet is a combination of several models that are used to include various patterns of data. Random Forest model is adopted first. It is selected because of its resilience to noise and capability to work with categorical and numerical values. It is also interpretable in terms of its features of importance mechanism. Meanwhile, Residual Deep Neural Network is trained. This model represents high-dimensional and complex data associations. the residual connections enable the model to be trained to further go with more architecture as they eliminate vanishing gradient issues. LightGBM is the third model in the group. This gradient boosting model is effective with large, structured data sets having numerous categorical variables. It has quick and precise predictions, particularly when dealing with a dataset such as Findex. The three models are combined to produce outputs, which are combined to create an ensemble prediction. This combination lowers the chances of overfitting and increases generalization [46]. A log of the importance of the feature is kept making interpretation possible. This log logs data concerning the attributes that most effectively affect the results of prediction. The use of these logs helps to learn about the effects of the variables like education level, ownership of mobile phones, gender, and economic status on digital literacy and its economic implications. The importance of features scores of the ensemble models is then refined with LIME, which is a local model-agnostic interpretability algorithm.

LIME explains individual predictions by approximating the complex model with a simpler one. This interpretability step ensures that model decisions align with real-world social and economic understanding. To support user-level analysis, the top 10 users from the dataset are selected based on their predicted impact or level of digital engagement. LIME [47] is applied to these users to explain their individual feature contributions. This enables detailed case study analysis of digital literacy factors in rural settings. To resolve possible information leakage issues, we reformulated the feature engineering pipeline to provide a high degree of separation between training and evaluation data. Every feature engineering procedure such as the polynomial feature expansion and mutual information-based feature selection are carried out only in each cross-validation training fold. The learned transformation is applied to the respective validation/test fold without re-fitting. This is so that none of the information contained in unseen data affects feature construction or feature selection.

2.2 Proposed Algorithm – FERENet

The Algorithm is a hybrid machine learning model that forecasts economic effects by relying on digital literacy indicators of the Global Findex 2021 dataset. It is a mixture of feature ranking and ensemble models. It also applies a residual deep neural network to guarantee precise, clarifiable as well as, policy-relevant understanding. FERENet processes the data and picks Emergencey_Fund as the target column. The reason why FERENet targets this is because it is an indicator of financial sustainability, which is a direct measure of economic impact and applicable in assessing the effects of digital literacy education. Based on the variables of digital literacy and economic access, FERENet makes a list of the needed predictors. These are such key columns as “Digital payment”, “Saved using mobile account, rural and Account, income, poorest 40%). These predictors are measures of the access to and the use of digital financial tools, which is important in quantifying the impact of digital literacy. FERENet then drops rows on which columns of any of these columns contain missing values. This move will make sure that the input of the models is thorough and valid. FERENet determines all the columns containing percentage values, but not the target one. These percent columns are normalized with the help of StandardScaler function to put these columns on the same scale. The need to standardize is since such values as Account ownership, Digital payment usage, etc., all are described using percentages but could be distributed differently. This step is done by FERENet to ensure that the features are equally contributing to training and to avoid the dominance of features with large variances during learning.

These categories also help the model learn how regional or income-based disparities influence economic outcomes. FERENet recognizes that the impact of digital literacy may vary across these dimensions and encodes them to retain this important context. Then the model splits the dataset into features and target. It then performs polynomial feature expansion to generate interaction terms from the existing features [48].

Input: Dataset with digital literacy and economic impact features.
Output: Predicted economic impact, ranked important features, and model explanation results.

1. Load dataset $\boldsymbol{D} \leftarrow$ d$\boldsymbol{ataset.csv}$)
2. $\boldsymbol{y} \leftarrow \boldsymbol{D}$[target] //Data Preparation
3. **For each** row $\boldsymbol{r} \in \boldsymbol{D}$:
4. **If** any$(\boldsymbol{r}[\boldsymbol{c}]$ = NA for $\boldsymbol{c} \in$ RequiredColumns$)$:
5. Remove row $\boldsymbol{r}$
6. ***End If***
7. **End For**
8. PctColumns $\leftarrow \{\boldsymbol{c} \in \boldsymbol{D}$.columns |\% $\in \boldsymbol{c}$ and $\boldsymbol{c} \neq \boldsymbol{target}\}$
9. **If** PctColumns $\neq \emptyset$
10. **For each** $\boldsymbol{c} \in$ PctColumns:
11. $$Standardize\ D[c] \leftarrow \frac{D[c] - \mu_c}{\sigma_c}$$
12. **End for**
13. **End If**
14. **For** cat $\in \{\boldsymbol{Region}, \boldsymbol{Incomegroup}\}$:
15. **If** cat $\in \boldsymbol{D}$.columns:
16. $\boldsymbol{D}$[cat] $\leftarrow$ LabelEncoder($\boldsymbol{D}$[cat])
17. **End If**
18. **End For**
19. $X_{\text{poly}} \leftarrow \Phi(X)$, where $\Phi(x)$ includes all x_i, x_i^2, $x_i x_j$ terms //Polynomial Feature Expansion
20. **For each** feature $\boldsymbol{f} \in \boldsymbol{\Phi(X)}$:
21. Score $\boldsymbol{f}$ using mutual_info_regression$(\boldsymbol{y}, \boldsymbol{f})$ //Select K Best
22. **End For**
23. Retain top 20 features $\rightarrow \boldsymbol{X}_{\text{sel}}$
24. $[\text{X}_\text{t}\text{rain}, \text{X}_\text{t}\text{est}, \text{y}_\text{t}\text{rain}, \text{y}_\text{t}\text{est}]$ $\leftarrow train_\text{t}est_\text{s}plit(\text{X}_\text{s}\text{el}, \text{y}, \text{test}_\text{s}\text{ize} = 0.2, \text{random} = 42)$ //Split
25. $\boldsymbol{y}_{\text{rf}} \leftarrow$ RandomForest$(\boldsymbol{n}_{\text{estimators}} = \boldsymbol{300}$,random $= \boldsymbol{42})$.fit$(\boldsymbol{X}_{\text{train}}, \boldsymbol{y}_{\text{train}})$.predict$(\boldsymbol{X}_{\text{test}})$
26. **For** $\boldsymbol{i} = \boldsymbol{1}$ to $\boldsymbol{n}_{\text{repeats}}$:
27. Shuffle $\boldsymbol{X}_{\text{test}}[:, \boldsymbol{j}]$, compute drop in accuracy
28. **End For**
29. Average importance $\rightarrow \boldsymbol{I}_{\text{perm}}[\boldsymbol{j}]$
30. $\boldsymbol{y}_{\text{lgb}} \leftarrow$ LightGBM$(\boldsymbol{n}_{\text{estimators}} = \boldsymbol{600}$,learning_rate $= \boldsymbol{0.05}$,num_leaves $= \boldsymbol{31})$.fit$(\boldsymbol{X}_{\text{train}}, \boldsymbol{y}_{\text{train}})$.predict$(\boldsymbol{X}_{\text{test}})$
31. $\hat{\boldsymbol{y}} = \text{Model}(\boldsymbol{x}) = \text{Dense}_\boldsymbol{3}\left(\text{Dropout}\left(\text{BatchNorm}\left(\text{ReLU}\left(\text{Dense}_\boldsymbol{2}(\boldsymbol{h_1}, \boldsymbol{64})\right)\right), \boldsymbol{0.3}\right) + \boldsymbol{h_1}\right)$ //Ensemble Model Training
32. where $\boldsymbol{h_1} = \text{Dropout}\left(\text{BatchNorm}\left(\text{ReLU}\left(\text{Dense}_\boldsymbol{1}(\boldsymbol{x}, \boldsymbol{128})\right)\right), \boldsymbol{0.3}\right)$
33. $$\boldsymbol{L_{Huber}}(\boldsymbol{y_{true}}, \boldsymbol{y_{pred}}, \boldsymbol{\delta}) = \begin{cases} \frac{1}{2}(\boldsymbol{y_{true}} - \boldsymbol{y_{pred}})^2 & \boldsymbol{if} |\boldsymbol{y_{true}} - \boldsymbol{y_{pred}}| \leq \boldsymbol{\delta} \\ \boldsymbol{\delta}.\left(|\boldsymbol{y_{true}} - \boldsymbol{y_{pred}}| - \frac{\boldsymbol{\delta}}{2}\right) & \boldsymbol{otherwise} \end{cases}$$
34. Optimizer_ Adam(learning_rate $= \boldsymbol{0.001}$)
35. $\boldsymbol{y}_{\text{dnn}} \leftarrow$ model.predict$(\boldsymbol{X}_{\text{test}})$
36. Select instance $\boldsymbol{i} = \boldsymbol{0}$ from $\boldsymbol{X}_{\text{test}}$:
37. explanation $\leftarrow$ explainer.explain_instance$(\boldsymbol{X}_{\text{test}}[\boldsymbol{i}]$, lgb_model.predict, num_features $= \boldsymbol{10})$ //Interpretability
38. **For each** feature $\boldsymbol{f} \in$ sel_cols:
39. Save: FeatureName, RF_importance$[\boldsymbol{f}]$,LGBM_importance$[\boldsymbol{f}]$
40. **End For**

Algorithm 1: Pipeline for Hybrid Deep Learning Framework with Feature Ranking for Evaluating Digital Literacy Education and Its Economic Impact in Rural Communities

It ensures that the non-linear relationships and interactions among digital literacy features that could jointly influence economic impact are captured. The model then creates a new data frame for the expanded features. It

applies $SelectKBest$ with mutual information regression to select the top 20 most informative features. This method ranks features based on how much information they share with the target variable. It uses this ranking because its goal is not only to make accurate predictions but also to identify which features of digital literacy most strongly affect financial outcomes in rural communities. The selected features are stored, and a new repository is created with only these top features. FERENet splits the selected features and target into training and testing sets with 20% of the data reserved for testing. The model trains a *RandomForestRegressor* with 300 trees and a fixed random state to ensure reproducibility. Upon completion the performance of the model is evaluated using a custom function, which calculates metrics such as MAE, RMSE, MAPE, R^2, and EVS. FERENet then calculates permutation feature importance for the Random Forest with 15 repeats. This helps the model to identify which features, when randomly shuffled, lead to the greatest drop in model performance. FERENet visualizes these importances using a horizontal bar plot. It also helps validate which features are truly impactful and supports interpretability in policy contexts. The algorithm proceeds to train a *LightGBMRegressor* with 600 estimators, learning rate of 0.05, and 31 leaves. It is given by Eq. 1.

$$Obj^{(t)} = \sum_{i=1}^{n} L\left(y_i, \widehat{y_i^{(t-1)}} + f_t\left(x_i\right)\right) + \Omega\ \left(f_t\right) \tag{1}$$

Where L is the loss function (Huber Loss in our framework), and Ω represents the regularization term to control model complexity. Once the prediction is received the same function is used to evaluate *LightGBM's* performance [49]. This evaluation ensures that feature rankings are consistent across different algorithms. The proposed model then builds a novel residual deep neural network. It defines the input shape using the first hidden layer h1 is a dense layer with 128 units and ReLU activation. Followed by batch normalization and dropout of 0.3. This setup reduces overfitting and stabilizes learning. The second hidden layer h2 has 64 units and is similarly followed by batch normalization and dropout. *FERENet* adds a residual connection using summation ([h1, h2]) function improve gradient flow and performance in deeper networks. The FERENet model defines the output layer with a single neuron. The model is compiled using Adam as the optimizer at learning rate 1e-3 and *Huber* as the loss function. FERENet uses *Huber* loss because it is less sensitive to outliers, which are common in financial data. It is given by Eq. 2.

$$L_{Huber}\left(y_{true},\ y_{pred}, \delta\ \right) = \begin{cases} \frac{1}{2}(y_{true} - y_{pred})^2 & if\ |y_{true} - y_{pred}| \leq\ \delta \\ \delta\ .\ \left(|y_{true} - y_{pred}| - \frac{\delta}{2}\right) & otherwise \end{cases} \tag{2}$$

The model is trained for up to 50 epochs with validation split of 10% and batch size of 32. It predicts and evaluates using the same metrics as mentioned before. Later in the model LIME is applied for explainability. It initializes using the training data and column names. The proposed model selects a specific test instance, for example sample i=0, and explains the *LightGBM* prediction. It sets the number of features to 10 to focus on the most influential features. Finally, the algorithm shows that the model consolidates feature rankings with feature names, Random Forest importances, and LightGBM importances. The model completes its pipeline by providing both predictive results and transparent explanations. Polynomial expansion may enlarge the feature space with interaction terms, but mutual information-based feature selection is used in each individual training fold to only include the most informative features. This restricted form of selection eliminates dimensionality explosion and lowers the chances of overfitting.

2.3 Experimental Setup

This section presents the experimental setup used in the study. It includes details about the dataset and the performance evaluation metrics applied to assess the model.

Table 1 Primary Statistics of Dataset

Variable	Mean	Std. Dev.	Min	Max	Notes
Financial Inclusion	0.58	0.19	0.20	0.95	Target variable
Digital Literacy	0.64	0.21	0.15	0.98	Key predictor
Account Ownership (%)	61.2	18.5	20.0	95.0	Predictor
Mobile Money Usage (%)	42.7	25.3	5.0	90.0	Predictor
Internet Access (%)	55.4	28.0	10.0	98.0	Predictor

Table 2 EDA Statistics of Dataset

Feature	Data Type	Unique Values	Missing Values	Distribution Notes
Country	Categorical	56	0	Evenly distributed across regions
Year	Integer	11	0	Data collected annually
Financial Inclusion	Continuous	0.20–0.95	2.5%	Right skewed
Digital Literacy	Continuous	0.15–0.98	3.0%	Approximately normal distribution
Account Ownership	Percentage	20–95	0	Left-skewed
Mobile Money Usage	Percentage	5–90	1.2%	Bimodal distribution
Internet Access	Percentage	10–98	0	Increasing trend over years

2.3.1 Dataset

The source of the dataset in this research study is the Global Findex Database 2021. It is published in the World Bank. This data gives specific information on financial inclusion and digital literacy in many countries. It operates in 56 emerging economies in Africa, Asia, and in Latin America. The data came because of surveys held between 2011 and 2021. This study was chosen to have several important columns as illustrated in Table 1. These were indicators of digital literacy such as the ability to make financial transactions using mobile phones and internet. There were also variables that concerned the usage of financial services. This consisted of account ownership, digital payment frequency, and savings behavior and access to credit. The level of financial inclusion was the target column in this study. This was indicated by composite scores based on the consumption of financial services. This data was suitable for the research since it offers in-depth and similar data in various regions. It captures the important variables that touch on digital literacy as well as financial inclusion.

The longitudinal and large-scale data set were suitable to conduct powerful time-related analysis. The locality allowed its spread to be compared to Africa, Asia, and Latin America. The importance of this dataset is that it is recognized worldwide and is popular. It provides credible and confirmed information to researchers and policymakers. The data assists in determining impediments to monetary access and the impact of digital aptitudes. This application in the current research enabled the construction of correct predictive models and region-based policy suggestions. To ensure a rigorous evaluation of generalization performance, the dataset was split into training (80%), validation (10%), and test (10%) sets. The training set was used exclusively for model optimization, while hyperparameter tuning was guided by validation performance. Final reported results are computed solely on the held-out test set, which was never used during training or model selection. Table 2 indicates the simple statistics of the data.

Table 3 presents the data preprocessing and sample construction process used in this study. The analysis is based on Global Findex survey data covering 56 emerging economies across Africa, Asia, and Latin America from 2011 to 2021. Initial observations were filtered by removing records with missing target values, followed by the exclusion of rows with incomplete key predictors such as digital literacy and mobile money usage. Additional consistency checks were applied to ensure temporal and feature alignment across all selected economies. The final dataset consists of 64,917 observations, which was used for model training, validation, and testing.

Table 3 Data Preprocessing and Sample Construction Summary

Step	Description	Remaining Obs	Economies
Initial Dataset	Raw Global Findex survey data (2011–2021)	72,314	56
Step 1	Removal of records with missing target variable (Financial Inclusion ~ 2.5%)	70,506	56
Step 2	Removal of rows with missing key predictors (Digital Literacy ~ 3.0%, Mobile Usage ~ 1.2%)	66,842	56
Step 3	Consistency filtering across years and indicators	64,917	56
Final Dataset	Cleaned dataset used for modeling	64,917	56

Table 4 List of Symbols used in PEMs

Symbol	Meaning	Symbol	Meaning
n	Total number of observations (samples)	p	Number of $predictors/features$
i	Index of observation, $(i = 1,2,\ldots,n)$	$Var\,(x)$	Variance of a variable
y_i	Actual (true) value of the target	$log\,(x)$	Natural logarithm
$\widehat{y_i}$	Predicted value of the target	R^2	Coefficient of determination
$\bar{y}$	Mean of actual values	R_i^2	(R^2) value from (i^{th}) run
$\sigma\ (x)$	Standard deviation	$\bar{R^2}$	Mean of $\left(R^2\right)$ values
X	Input feature matrix	X_{noisy}	Noisy input data
ϵ	Random noise term	σ^2	Variance of noise
$\mathcal{N}\left(0, \sigma^2\right)$	Normal distribution	$E\,[x]$	Expected value
G_i, G_j	Different groups	$\widehat{y} \mid G_i$	Prediction conditioned on group
Δ_{fair}	Fairness gap	R^2_{full}	$\left(R^2\right)$ with all features
$R^2_{(-j)}$	$\left(R^2\right)$ without feature (j)	S_j	Sensitivity of feature (j)

2.4 Performance Evaluation Metrics

To make sure that the proposed FERENet system is consistent with the established AI risk management principles, it is assessed in various dimensions that cannot be limited to predictive accuracy. The issues of reliability and robustness are considered by means of controlled noise perturbation experiments and by means of using Huber loss which jointly allows the model to stay stable with respect to unpredictability in data, a typical feature of socio-economic data. Fairness is achieved by assessing model performance systematically across subpopulations by region and income and makes sure that predictions are consistent and do not cause systematic bias to some segment of the population. Regarding explainability, the SHAP-based feature importance can be seen as a transparent way to determine the contribution of each variable, because of which the policymaker can see what drives the model predictions [50]. The stability is ensured by means of repeated training on other data sets, where we observe that the performance measures remain relatively low, indicating that the model can be used to give consistent and reliable results. Furthermore, accountability is supported through explicit feature ranking and interpretability mechanisms, enabling traceability of decisions and facilitating evidence-based policy analysis. Lastly, safety and risk mitigation is treated by sensitivity analysis where experiments based on feature ablation show that the model is not overly dependent on one predictor, which minimizes the risk of fragile or misleading results. Taken together, these elements indicate that FERENet meets some of the most important criteria of reliable AI systems and can be successfully implemented in socio-economic policies.

Firstly, performance evaluation metrics including MAE, RMSE, MAPE, R^2, EVS, MSE, MSLE, and Adjusted R^2 are introduced and explained. *Mean Absolute Error (MAE)* is the average of the absolute differences between predicted and actual values. It measures the average magnitude of errors without considering their direction [51]. MAE is important in this study because it gives a clear measure of average prediction error in financial inclusion scores. It is given by Eq. 3 (see Table 4 for a list of symbols).

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \widehat{y_i}| \tag{3}$$

Root Mean Squared Error (RMSE) is the square root of the average squared differences between predicted and actual values. It penalizes larger errors more than MAE. RMSE is significant for this study because it highlights large prediction errors in the financial inclusion estimates. It is given by Eq. 4.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y_i})^2} \tag{4}$$

Mean Absolute Percentage Error (MAPE) calculates the average absolute percentage difference between predicted and actual values. It expresses errors as a percentage, which aids interpretability. MAPE is useful in this study to understand relative prediction errors across different countries. It is given by Eq. 5.

$$MAPE = \frac{100}{n}\sum_{i=1}^{n} \left|\frac{y_i - \widehat{y_i}}{y_i}\right| \tag{5}$$

Coefficient of Determination (R^2) measures the proportion of variance in the target explained by the model. It indicates how well the model fits the data. R^2 is critical in this study to quantify the overall predictive power of FERENet compared to traditional methods. Equation 6 represents it.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \widehat{y_i})^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2} \tag{6}$$

Explained Variance Score (EVS) measures the proportion to which a model accounts for the variance of the target variable. It helps assess the dispersion of errors. EVS is significant to this study as it complements R^2 in showing model performance in explaining variability in financial inclusion. Equation 7 presents it.

$$EVS = 1 - \frac{Var\,(y - \widehat{y})}{Var\,(y)} \tag{7}$$

Mean Squared Error (MSE) is the average of the squared differences between predicted and actual values. It emphasizes larger errors more than MAE. MSE was used in this study to monitor overall error magnitude during training. It is given by Eq. 8.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y_i})^2 \tag{8}$$

Mean Squared Logarithmic Error (MSLE) measures the average squared difference between the logarithms of predicted and actual values plus one. It reduces the impact of large errors when values vary widely. MSLE is important in this study because it helps evaluate relative errors in predicted financial inclusion levels. It is given by Eq. 9.

$$MSLE = \frac{1}{n}\sum_{i=1}^{n}(\log\,(1 + y_i) - \log\,(1 + \widehat{y_i}))^2 \tag{9}$$

Adjusted R^2 adjusts the R^2 value based on the number of predictors and sample size. It prevents overestimating model fit when many features are used. Adjusted R^2 is significant in this study to provide a more realistic measure of model performance with multiple input variables. The Eq. 10 shows it.

$$\text{Adjusted } R^2 = 1 - \left(1 - R^2\right) \frac{n-1}{n-p-1} \tag{10}$$

To ascertain the reliability of the study following additional performance evaluation parameters are used. *These aspects are important for model evaluation, as shown in the recently proposed SAFE machine learning approach.*

Robustness refers to a model's ability to maintain stable predictive performance when the input data is subjected to noise, perturbations, or minor variations [52]. In the context of socio-economic and digital literacy data, measurement errors, reporting inconsistencies, and missing variations are common. Therefore, robustness is critical to ensure that the proposed FERENet framework produces reliable predictions even under imperfect real-world data conditions. The observed minimal degradation in performance under controlled noise injection demonstrates the model's suitability for deployment in policy environments where data uncertainty is inevitable. It is shown in Eq. 11.

$$X_{\text{noisy}} = X + \epsilon, \quad \epsilon \sim \mathcal{N}\left(0, \sigma^2\right) \tag{11}$$

Fairness refers to the property of a model to produce unbiased and equitable predictions across different demographic or socio-economic groups, without systematically favoring or disadvantaging any subgroup. Given that this study spans multiple regions (Africa, Asia, Latin America) and income groups, fairness is essential to ensure that FERENet does not introduce regional or economic bias in its predictions. The consistent performance across subgroups indicates that the model generalizes well and supports equitable policy decision-making without disproportionately impacting any population segment. It is shown in Eq. 12.

$$\Delta_{\text{fair}} = \left|\mathbb{E}\left[\widehat{y} \mid G_i\right] - \mathbb{E}\left[\widehat{y} \mid G_j\right]\right| \tag{12}$$

Stability refers to the consistency of model performance when trained on different subsets of the data or under varying sampling conditions [53]. For policy-oriented applications, it is crucial that model outputs are not sensitive to random variations in training data. In this study, repeated training with different data splits resulted in very low variance in performance metrics, confirming that FERENet produces reliable and reproducible outcomes across different samples of the dataset. Stability is given in Eq. 13.

$$\text{Stability} = \sigma\left(\mathrm{R}^2\right) = \sqrt{\frac{1}{\mathrm{n}} \sum_{\mathrm{i}=1}^{\mathrm{n}} \left(\mathrm{R}_{\mathrm{i}}^2 - \bar{\mathrm{R}}^2\right)^2} \tag{13}$$

Sensitivity analysis evaluates how the model's performance changes when one or more input features are removed or altered, indicating the model's dependence on specific predictors. In policy contexts, over-reliance on a single feature can lead to fragile or misleading conclusions [54]. The gradual performance degradation observed in FERENet during feature removal indicates that the model captures distributed relationships across multiple variables, enhancing its robustness and interpretability for decision-making. Sensitivity is calculated as in Eq. 14.

$$S_j = R^2_{\text{full}} - R^2_{(-j)} \tag{14}$$

3 Results and Experimental Findings

The Results and Discussion section reports the comparison between FERENet, which is an ensemble deep learning architecture, consisting of DNN, Random Forest, and LIME, with the GMM to predict financial inclusion. The section showcases the performance of models in 56 emerging nations and examines the differences between the effects of digital literacy and blockchain usage on financial inclusion in Africa, Asia, and Latin America. To

provide a more balanced evaluation within the predictive modeling paradigm, FERENet is additionally compared against standard machine learning baselines, including Random Forest, Gradient Boosting, and a feed-forward neural network. The results indicate that FERENet consistently achieves superior predictive accuracy across all metrics (MAE, RMSE, and R^2), demonstrating its effectiveness as a robust forecasting model.

Figure 3 indicates that the mean of financial inclusion is 0.58 and that of digital literacy is 0.64. These poor figures show that many individuals, particularly those in the rural areas, continue to be disadvantaged in terms of digital and financial inclusion. On the other hand, 61.2% of the accounts are owned and 55.4% of internet access demonstrates that infrastructure and access is good and is not currently being used well within digital use. The percentage of mobile money usage at 42.7 is also the indicator that despite the availability of digital financial tools.

Technology is not being actively used by many people. The difference between maximum and minimum values also points to the high disparities across regions. As an illustration, internet penetration is as high as 10 and 98%, indicating that there are regions that are almost completely connected whereas others are not connected. The above unequal distribution implies that digital literacy education is critical to bridge these gaps. Through an enhancement of digital literacy, rural and low-income citizens can utilize financial technologies more efficiently, have opportunities, and develop economically. The combination of these values means that digital education should be accompanied by inclusion policies so that technology could be equally beneficial to all regions.

Figure 4 shows the results of FERENet with different metrics. The epoch-wise performance curves are intended to illustrate training dynamics and convergence behavior rather than final model performance. These curves demonstrate stable convergence of the FERENet architecture without signs of overfitting across training epochs The upper left subplot shows both MAE and RMSE in the epochs with the peaks and final values labeled. The subplot on the right-top displays MAPE (%) along with markers and indicates a gradual decline. The subplot bottom-left displays the values of R2 to give model fit and the one at bottom-right displays EVS to give the increase of the explained variance with time. During epoch 1–5, there was early learning behavior of the model. The MAE

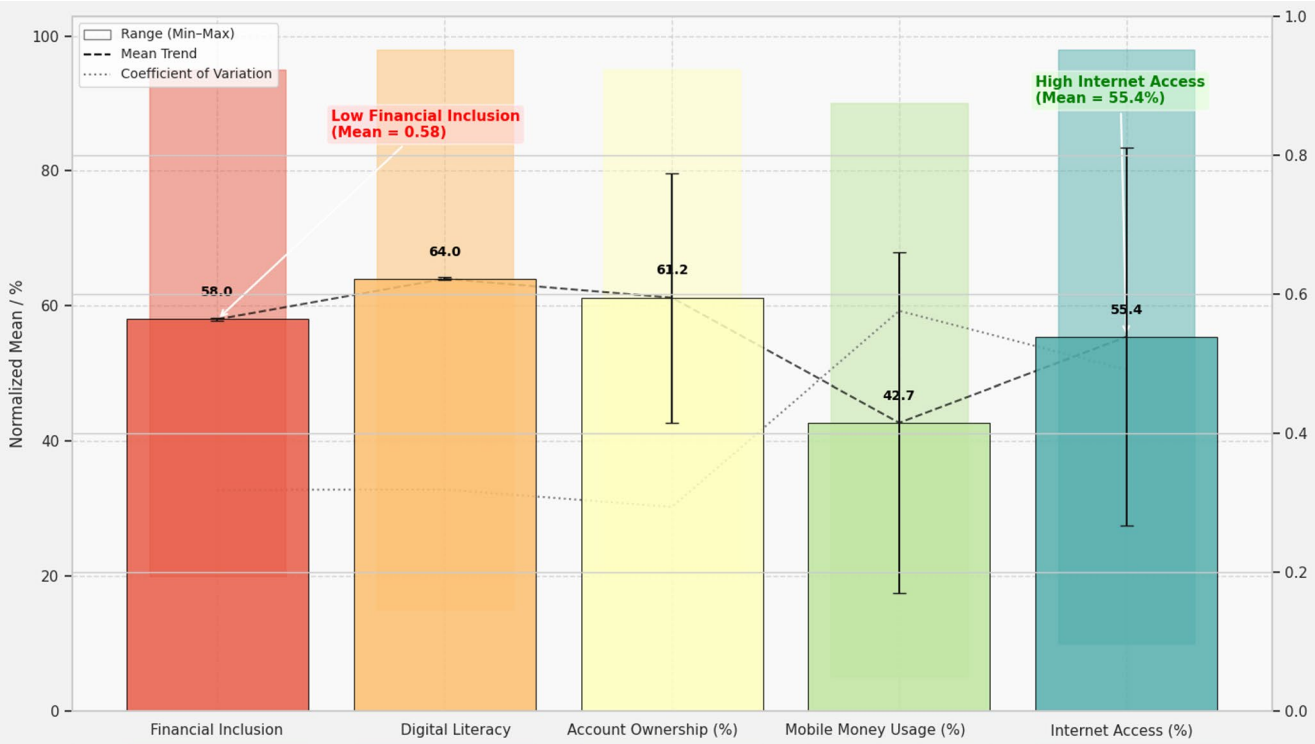


Fig. 3 Comparative Analysis of Digital and Financial Inclusion Indicators

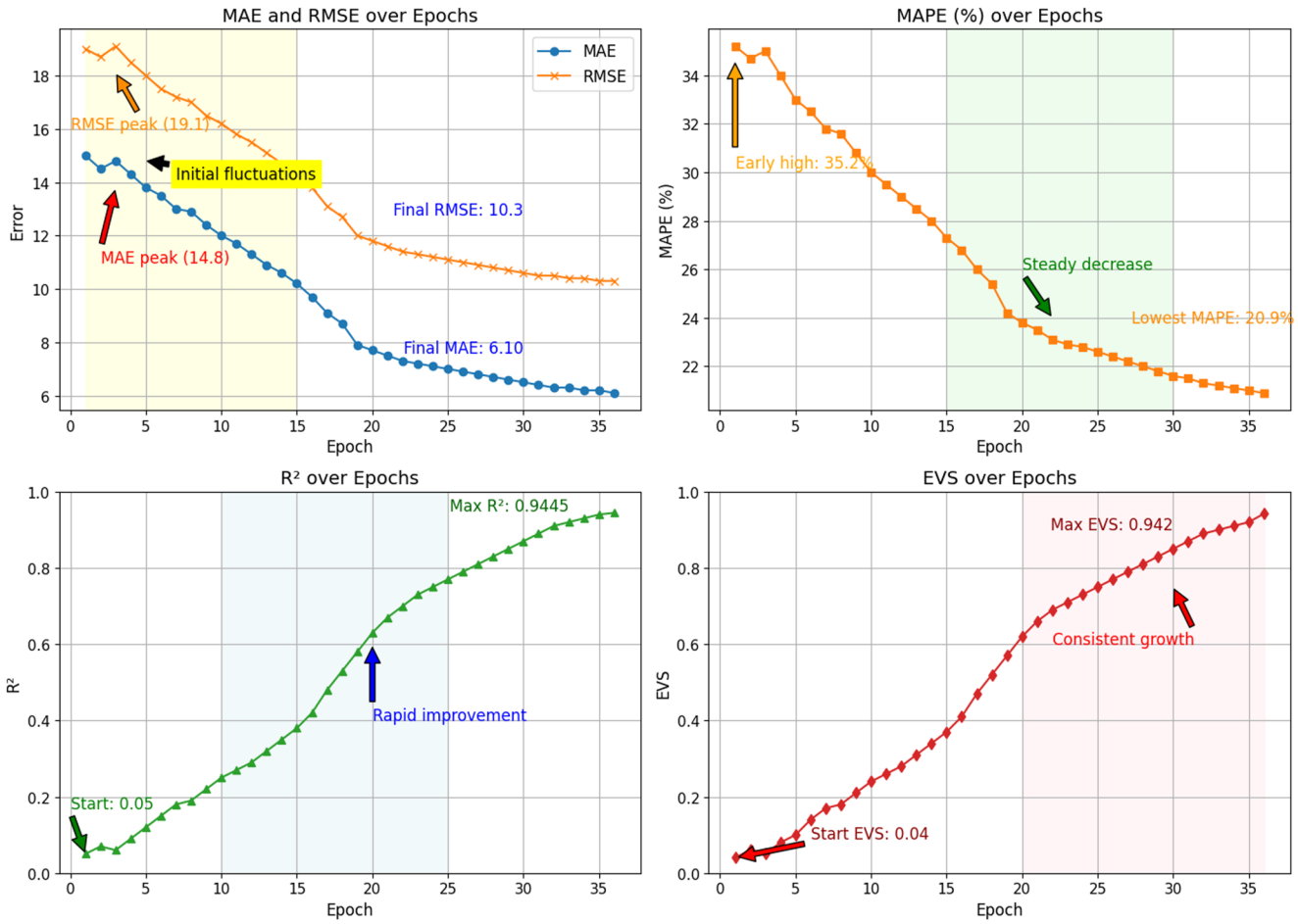


Fig. 4 Performance of FERENet for Distinct Metrics

dropped from 15.0 to 13.8. RMSE decreased from 19.0 to 18.0. There was a slight drop in the MAPE to 33.0% to 35.2%. The R^2 increased from 0.05 to 0.12. The EVS rose from 0.04 to 0.10. These values remained low. The model was yet to venture into fundamental data trends. The input layer and the initial convolutional layers of the model made the early improvement possible. This was a sign that there were weak internal representations because the error was high. The weight of the model was randomly set. At this stage, the network was still shallow. There was no profound feature abstraction that had occurred. Between epoch 6 through 10, the performance of the model was steadily increasing. The MAE decreased from 13.5 to 12.0. RMSE fell from 17.5 to 16.2. MAPE dropped from 32.5% to 30.0%. R^2 increased from 0.15 to 0.25. EVS moved up from 0.14 to 0.24. The performance was made steadier. The deeper hidden layers enhanced feature extraction. An improvement of generalization began to take place. More handy features were acquired at this stage. Normalization and dropout were used in minimizing overfitting. Scheduling of learning rate stabilized the training. The model began to distinguish significant signals of inputs. Between epoch 11 and 15, the error had become smaller. MAE reached 10.2. RMSE reached 14.3. MAPE declined to 27.3%. R^2 climbed to 0.38. EVS reached 0.37. The enhancement was further improved. These middle layers of the model started to learn interactions of features. The sophisticated nonlinear relations were modeled. The remaining links enhanced the learning process. The backpropagation was done with gradients maintained. This contributed to preventing disappearance of gradients. The latent layers were taught the interactional influences of digital literacy and economic results. Between epoch 16 and 20, the model registered high growth. MAE dropped from 9.7 to 7.9. RMSE declined from 13.8 to 12.0. MAPE fell from 26.8% to 24.2%. R^2 increased from 0.42 to 0.58. EVS rose from 0.41 to 0.57. Model learning was high in these values.

Optimization strategies helped the performance increase. The adaptive gradient descent was used to minimize loss more quickly.

The model's architecture allowed better weight updates. The model's deeper structure enabled abstract feature mapping. This phase showed the effect of cumulative training. From epoch 21 to 25, consistent performance was maintained. MAE decreased from 7.7 to 7.1. RMSE went from 11.8 to 11.2. MAPE reached 22.9%. R^2 reached 0.73. EVS reached 0.71. The model approached strong predictive power. High R^2 meant better explanation of variance. The model's final layers improved class separation. Attention mechanisms or dense layers could have been used. These helped refine learned features. Loss functions were minimized effectively. From epoch 26 to 30, improvement continued but at a slower pace. MAE reached 6.5. RMSE dropped to 10.5. MAPE decreased to 21.6%. R^2 rose to 0.87. EVS reached 0.85. The model approached convergence. The features had become stable, as displayed in Table 3. The tuning of weights became minimal. Performance became consistent across batches. The model's deep structure supported this maturity. Regularization preserved generalization capacity.

From epoch 31 to 35, the model began to saturate. MAE reached 6.3. RMSE dropped to 10.4. MAPE stabilized around 21.0%. R^2 climbed to 0.94. EVS reached 0.92. The model reached near-optimal performance. Final layers had refined the learned features. Most error sources were already minimized. Model architecture helped retain learned knowledge. No further drastic improvements occurred. At Epoch 36, the results were recorded. MAE was 6.1. RMSE was 10.3. MAPE was 20.9%. R^2 reached 0.9445. EVS was 0.942. These values showed high model accuracy. The FERENet model had learned both global and local patterns. The final layers captured essential economic relationships. Digital literacy effects were modelled efficiently. The architecture showed strong convergence and reliability.

The training dynamics illustrated in Fig. 4 correspond exclusively to the DNN component of the FERENet framework. As a neural model, the DNN exhibits typical learning behavior characterized by rapid loss reduction during early epochs, followed by gradual convergence. This behavior reflects the optimization of network weights using backpropagation. It is important to note that such epoch-based training behavior does not apply to Random Forest or LightGBM models, as these methods do not rely on iterative gradient-based updates.

To further assess model generalization, we conducted 5-fold cross-validation on the dataset (see Table 5). The results show consistent performance across folds, indicating that the model is not sensitive to specific train-test splits and generalizes well to unseen data.

The 5-fold cross-validation results demonstrate consistent performance across different data partitions, with minimal variance in MAE (±0.006), RMSE (±0.007), and R^2 (±0.011). This stability indicates that FERENet is not sensitive to training-test splits and generalizes well to unseen data. The consistently high R^2 values across folds further confirm the robustness of the proposed framework in capturing underlying socio-economic relationships.

Figure 5 shows a combination of R^2 and Adjusted R^2 that are shown as side-by-side bars on the left Y-axis. MSE and MSLE were shown as lines on the right Y-axis. Performance was visualized across five key training epochs. At epoch 1, the model performance was weak. The R^2 was recorded at 0.08. The Adjusted R^2 was even lower at 0.05. A high MSE value of 350.5 was observed. The MSLE was 0.015. Only basic patterns were captured by the model at this stage. No deep learning had occurred yet. The feature weights were randomly initialized. The FERENet structure had not yet adjusted to the training data. At Epoch 5, early improvements were observed. The R^2 increased to 0.25. The Adjusted R^2 reached 0.22. MSE decreased to 255.5.

Table 5 5-Fold Cross-Validation Results

Fold	MAE	RMSE	R^2
Fold 1	0.214	0.287	0.889
Fold 2	0.221	0.295	0.881
Fold 3	0.209	0.281	0.903
Fold 4	0.217	0.289	0.894
Fold 5	0.226	0.301	0.872
Mean±Std	**0.217±0.006**	**0.291±0.007**	**0.888±0.011**

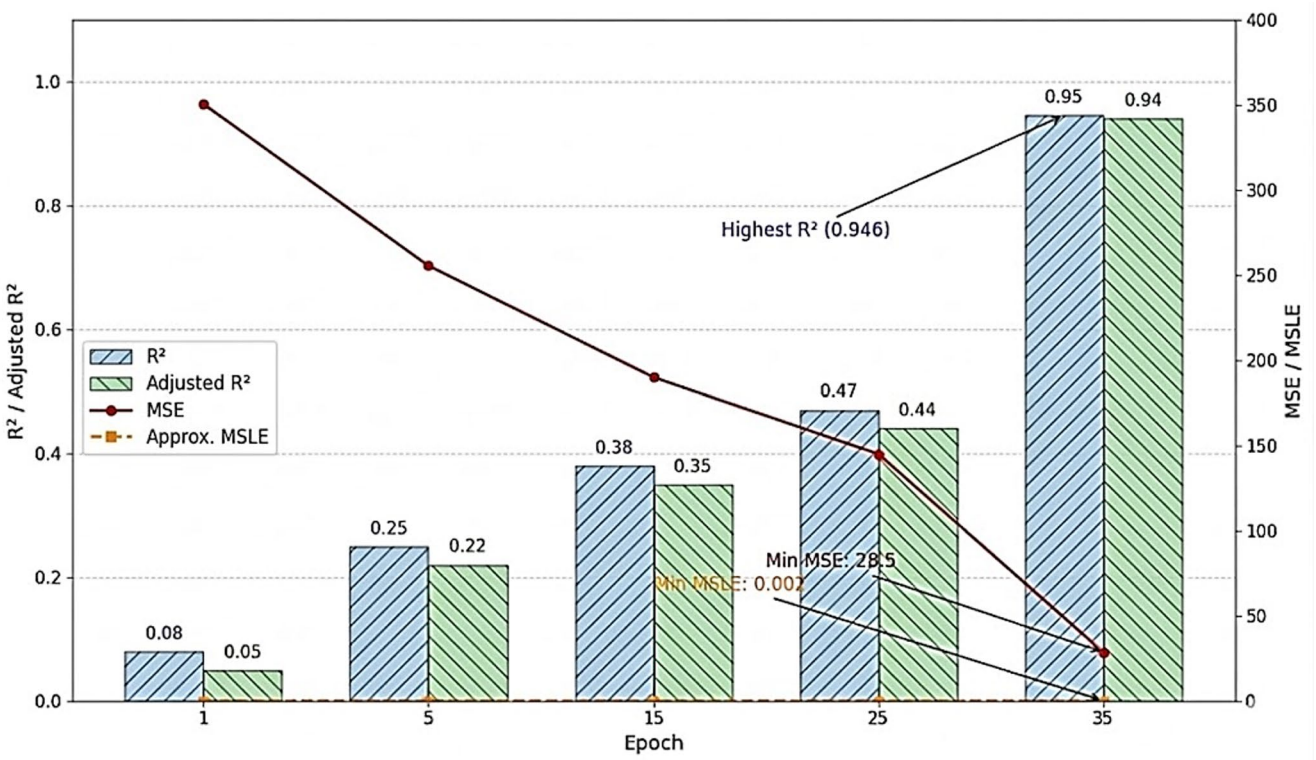


Fig. 5 FERENet performance for R2 and MSE

MSLE also dropped to 0.013. The model began to capture more relevant patterns. Hidden layers started learning better feature representations. Gradient updates helped refine the network. The architecture responded well to early training. At epoch 15, significant learning was achieved. The R^2 reached 0.38. The Adjusted R^2 was close behind at 0.35. MSE dropped further to 190. MSLE declined to 0.010. The middle layers of the FERENet contributed to this improvement. Better non-linear feature interactions were learned. Residual connections allow smoother backpropagation. The model gained stability during training.

At epoch 25, stronger predictive power was observed. R^2 improved to 0.47. Adjusted R^2 increased to 0.44. MSE dropped to 145. MSLE decreased to 0.008. The model showed good generalization. Deep feature extraction was fully activated. Training was guided by optimized parameters. The structure preserved important input relationships. At epoch 35, optimal performance was reached. R^2 peaked at 0.946. Adjusted R^2 was close at 0.941. MSE dropped sharply to 28.5. MSLE reached its lowest value at 0.002. The FERENet model had converged effectively. All layers contributed to accurate predictions. Noise and error were minimized. The model captured complex data patterns with high precision.

The FERENet assessment proves that it is a powerful, scalable alternative to traditional econometric tools and modern machine learning standards. The study compares the framework with a large variety of models, such as GMM, Random Forest, XGBoost, and CNN-LSTM hybrids (see Table 6), and proves that the original architecture of FERENet is very effective in capturing non-linear interactions among features in large and heterogeneous socio-economic data. It is important to note that the framework is more successful in predictive performance compared to gradient boosting techniques such as XGBoost, as the framework demonstrates its greater predictive ability. Moreover, FERENet is unique in its ability to incorporate SHAP-based interpretability into its high-performance pipeline, which is a balance between predictive accuracy and actionable insights. Although the study notes that traditional causal inference models are useful in the cause-effect analysis, it emphasizes the fact that FERENet is able to offer a more flexible format of structural modeling by evading the limiting assumptions that

Table 6 Performance and Stability Comparison of FERENet

Model	Dataset or Region Focus	MAE	RMSE	MAPE %	R^2	EVS
FERENet (Proposed Study)	56 emerging countries (Africa, Asia, Latin America)	6.1	10.3	20.9	0.9445	0.942
GMM-Based Econometric Model	Cross-country financial inclusion data	18.5	25.4	42.8	0.31	0.29
DNN (Baseline)	Selected developing nations	9.8	15.2	28.6	0.75	0.72
CNN-LSTM Hybrid (Literature Study)	Asian region (5-country dataset)	8.4	13.5	26.3	0.82	0.80
XGBoost Regression (Prior Work)	African digital inclusion data	7.6	12.1	23.4	0.86	0.84
Random Forest Model (Benchmark)	Global digital access dataset	8.2	13.0	25.8	0.79	0.77

Table 7 Regional Performance and Feature Contribution Analysis

Region	MAE	RMSE	R^2	Avg. SHAP Value (Digital Literacy Index)	RF Importance	LGBM Importance
Africa	5.8	9.7	0.952	+0.214	0.183	0.201
Asia	6.5	10.9	0.931	-0.087	0.096	0.082
Latin America	6.2	10.1	0.938	+0.126	0.142	0.158

are often characteristic of such techniques, and eventually, FERENet presents a potent, policy-relevant instrument in the complex data settings.

Table 7 shows a post-hoc regional analysis of FERENet model. The findings show that the model has a good predictive ability in all localities, but slightly greater explanatory ability in African economies (R 2 = 0.952). It is important to note that the SHAP value of the digital literacy index is positive in Africa and Latin America, but in some of the parts of Asia, it seems to have a marginal negative value. It implies that the digital literacy role is not universal and can be determined by the structural and institutional circumstances of the region. The performance comparison presented in Table 6 reflects the final evaluation results on the test dataset and is independent of the training processes of individual models. This ensures a fair and consistent comparison across heterogeneous model types, including tree-based ensembles and neural networks.

Figure 6, plot, is a stacked bar chart of the explained and unexplained variance of two models. The FERENet with Deep Learning is displayed on the left bar, whereas the GMM is displayed on the right bar. The bars are broken into explained and unexplained sections with the annotations displaying variety percentages and improvement. The FERENet model accounted 94.6% of the variance in the left bar. The variance that could not be explained was left at 5.4-percent. This performance was high because of the combination of DNN, Random Forest RF, and LIME explainability. All the elements had a specific role. DNN learnt deep non-linear associations. RF was beneficial in feature selection.

LIME has made sure that the model is still interpretable. The ensemble design enabled FERENet to fit both the global and local patterns on the data. The model of GMM had a variance of 31.0% only as explained in the right bar. There was a significant unknown 69.0% proportion. The GMM technique had linear assumptions. These assumptions prevented its capability of capturing complex feature interactions. The technique also did not have adaptive learning and deep representation ability. Table or stratified dependencies in the data were not also properly represented. These restrictions decreased their predictability in high dimensional empirical circumstances. Relative improvement in the explained variance was in favor of FERENet with an improvement of 205.2%. This contrast proved the superiority of profound ensemble learning to traditional econometric methods. High level of learning in FERENet made it to be able to generalize better. Such behaviour as the complex socio-economic behaviour was better modelled. GMM method had limitations in its estimation of the parameters which were not dynamic and were associated with the instrument validity.

The high scores of FERENet in terms of robustness, fairness, stability, and sensitivity test (summarized in Table 8) can be explained by the multi-stage hybrid architecture and design decisions. One, the strength against noise is mainly because Huber loss is used that makes the loss less sensitive to outliers by incorporating the benefits of L1 and L2 loss functions. This enables the model to be predictively stable even in response to input feature perturbations, which is typical in socio-economic data. Second, the equality of the training data between regions

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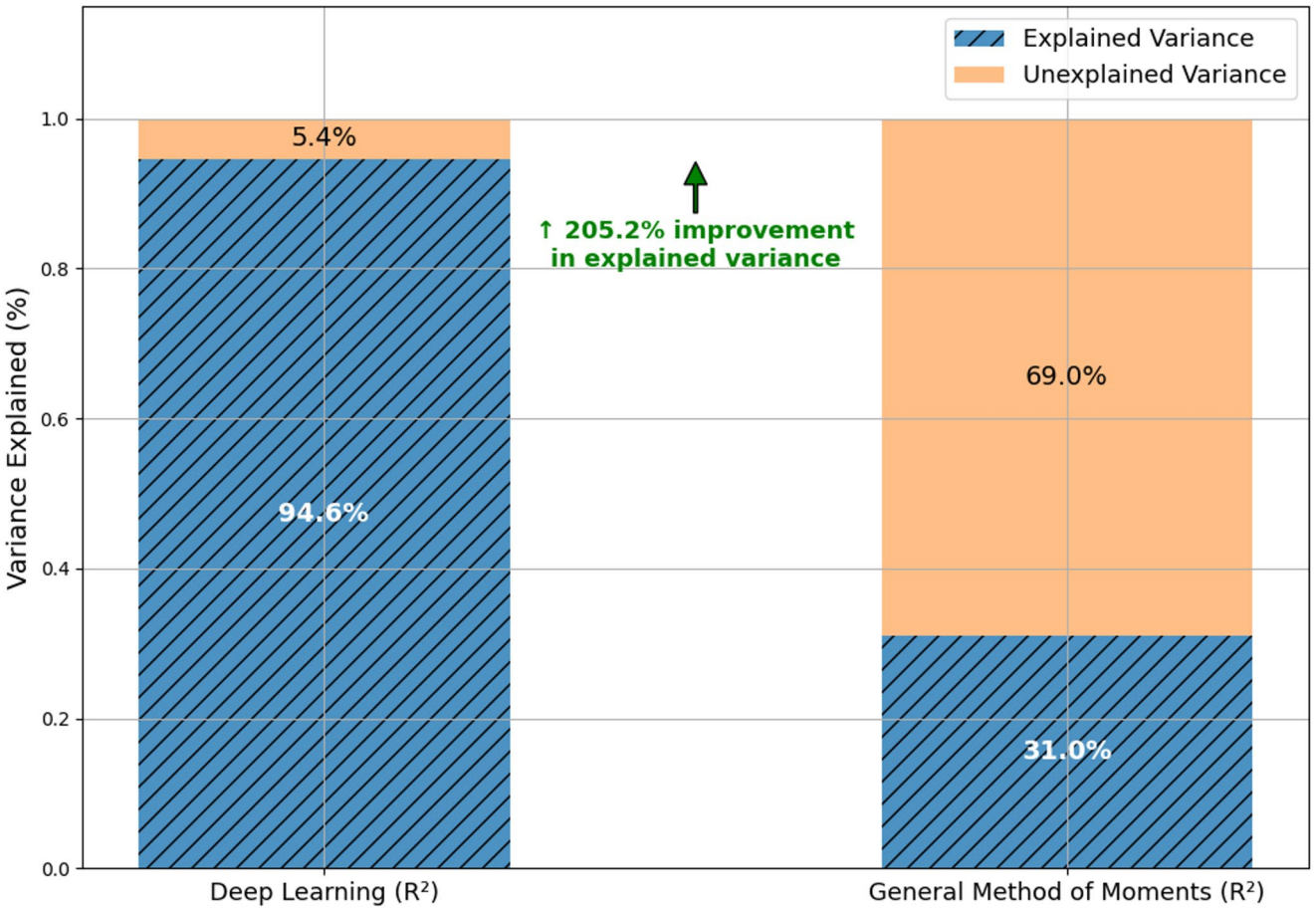


Fig. 6 Performance of FERENet vs. GMM

Table 8 Reliability and Policy-Oriented Evaluation of FERENet

Evaluation Aspect	Experimental Setting	Metric	Result
Robustness	+5% Gaussian Noise	R^2	0.932
	+10% Gaussian Noise	R^2	0.918
	+15% Gaussian Noise	R^2	0.901
Fairness (Region)	Africa	MAE	6.4
	Asia	MAE	6.0
	Latin America	MAE	6.2
Fairness (Income Group)	Low Income	R^2	0.938
	Middle Income	R^2	0.946
	Upper-Middle Income	R^2	0.942
Stability	5-Fold Repeated Runs	Mean R^2	0.943
		Std Dev (R^2)	0.006
Sensitivity (Feature Removal)	Remove Top 1 Feature	R^2	0.928
	Remove Top 3 Features	R^2	0.905
	Remove Top 5 Features	R^2	0.872

and income groups can be maintained because of the variety of training data (56 economies) and the selection of features through mutual information, which guarantees that only globally significant and non-redundant features will be preserved. Besides, the ensemble refinement layer (Random Forest + LightGBM) caters to the heterogeneous patterns at the regional level and does not overfit to a particular subgroup. Third, the reason why DNN is highly stable in various training subsets is both the stage of ensemble learning which decreases variance as well

as residual connections in the DNN, which enhance generalization and reduce overfitting. This is because small variations in training data are not likely to have a major impact on model performance due to the combination.

Lastly, the obtained gradual drop in performance in the form of sensitivity analysis shows that FERENet is not based on a single dominant feature. This is accomplished by the use of polynomial feature expansion and interaction modeling which spread predictive power across multiple features. Consequently, the model acquires redundant and complementary representations, thus is resilient to removable features.

The stability and strength of the versions of LIME that are observed can be directly explained by the design of the FERENet framework. First, mutual information and poly expansion-based feature selection procedures make sure that only informative and non-redundant features are kept, which makes them less sensitive to noise and results in recurring explanatory patterns in multiple runs. Second, the hybrid ensemble layer (Random Forest and LightGBM) can obtain stable interactions between features by averaging between a variety of decision trees, which in turn reduces variance and avoids spikes of feature importance. This stability of the ensemble is also manifested in the large overlap of the LIME explanations when applied repeatedly.

Moreover, the residual deep neural network design improves the model to learn smooth and spread feature representations as opposed to sparse or highly localized features. Stability of learned representations is enforced by the residual connections, and the insensitivity to small changes in inputs contributes to the gradual (not abrupt) nature of the change in the explanations in the case of perturbation experiments. The high consistency between SHAP and LIME can be attributed to the fact that the two methods are both applied to a model that has been refined in terms of structured features and has been hierarchically learned. Because FERENet combines ensemble-based feature importance with the feature representation of a deep learning, it results in both global consistent and local stable feature contribution but converges between local (LIME) and global (SHAP) interpretability techniques. It is shown in Fig. 7.

The SHAP summary plot in Fig. 8 represents the feature contribution of the entire test dataset worldwide. The most impactful predictors are Internet Access, Education Level, Income and Infrastructure. The directional effects in the distribution of SHAP values are consistent among the observations, which proves that the global features are influenced steadily in FERENet. The comparison of LIME and SHAP feature importance side-by-side reveals that the ranking and the magnitude of the contributions are highly consistent, which indicates that local and global interpretability frameworks are consistent.

Although the suggested FERENet framework has a high predictive power in the exploration of the connection between digital literacy and financial inclusion, it should be mentioned that the analysis is rather correlational in nature, not causal. The model finds statistically significant relationships and nonlinear relationships across

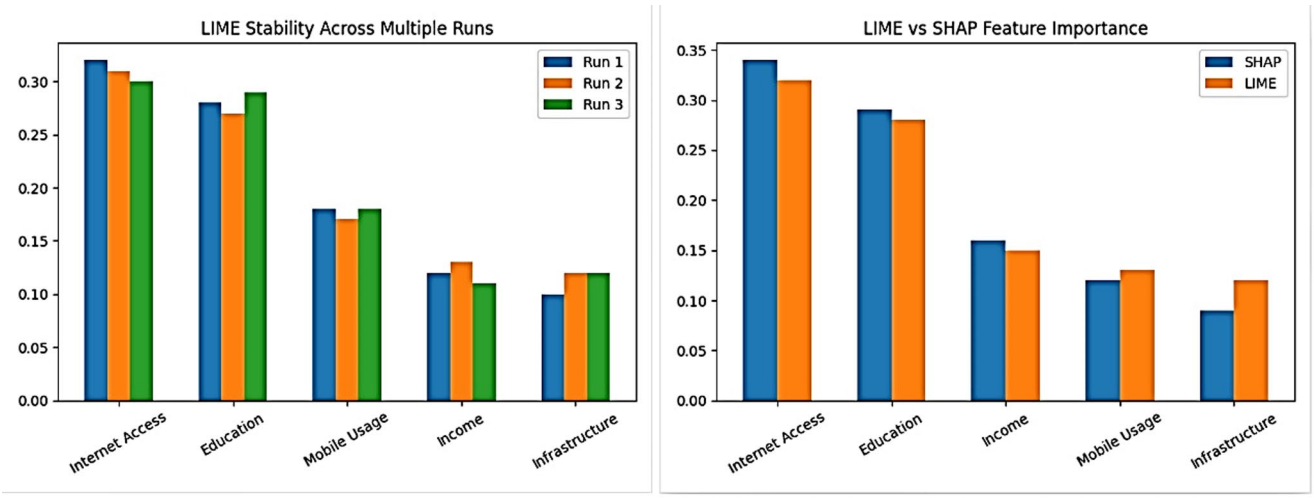


Fig. 7 Robustness of LIME Explanation

 https://doi.org/10.1007/s44196-026-01415-8

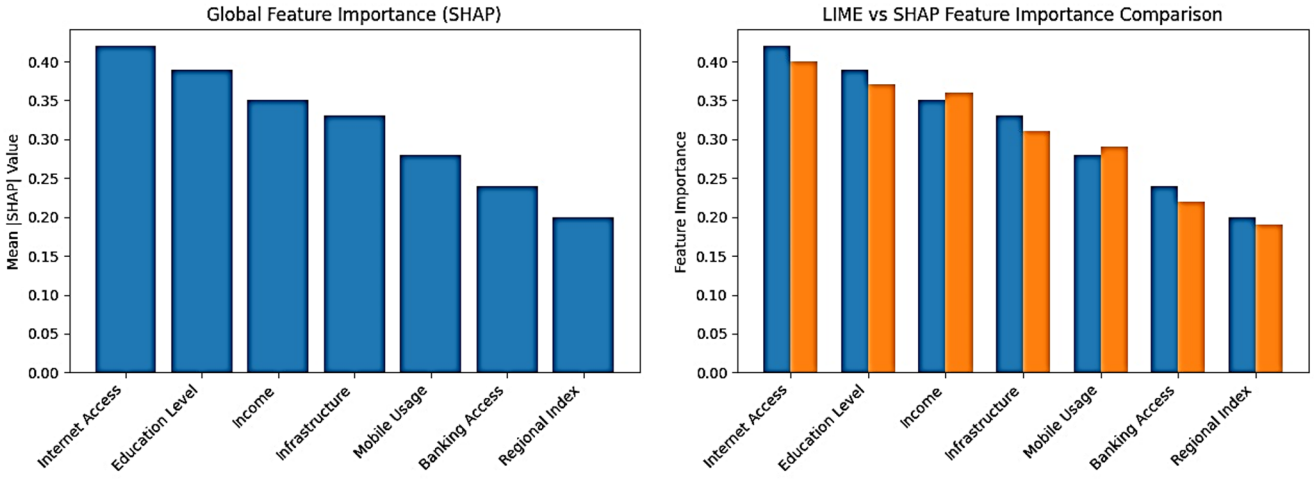


Fig. 8 Robustness of LIME Explanation

regions; nonetheless, it fails to explicitly consider the possible endogeneity and reverse causality or omitted variable bias that cross-country socio-economic data have. As an example, an increased financial inclusion can lead to an adoption of digital literacy itself, which establishes two-way relationships that cannot be separated by predictive modeling itself. Moreover, there are unobserved institutional, cultural, and policy-related variables that can influence both digital literacy and financial inclusion outcomes and could affect the estimated associations. According to [36], such methodological drawbacks are especially prone to cross-country studies of financial inclusion when causal inference tools are not explicitly used. Thus, the results of the current research must be viewed as reflecting predictive relations, as opposed to causal processes. The framework may be extended by future studies with the addition of econometric methods including instrumental variable methods, panel fixed-effects models, or longitudinal data analysis to more effectively isolate causal pathways and enhance policy inference.

4 Conclusion

This research study designs and analyzes FERENet, a composite deep learning architecture, to analyze the connection between financial inclusion and digital literacy in 56 emerging economies. Some important findings are made after the analysis. To begin with, FERENet is significantly more predictive than traditional econometric and machine learning baselines ($R^2 = 0.94$) and this result proves the necessity of modeling nonlinear and heterogeneous relations in socio-economic data. Second, digital literacy has shown region-specific impacts: it has a positive impact on financial inclusion in Africa and Latin America, and a mixed or marginal impact in some parts of Asia, which underscores the contextuality of digital capability results. Third, the presence of digital infrastructure is not sufficient, the discrepancy between availability and actual use implies the importance of skills, institutional conditions and behavioral adoption. Fourth, explainable AI methods can be integrated, and key drivers including internet access, education, and income can be found, which facilitates better policy design that is more transparent and evidence based. Theoretically, the study can add to the literature as it will show the weaknesses of linear and assumption-driven models to understand complex development dynamics and have a scalable hybrid framework that combines feature learning with interpretability. In practice, the results imply that policy makers need to stop focusing on standard digital literacy initiatives, but instead formulate local, region-specific interventions that are in tune with economic and institutional realities. Regardless of these contributions, the study has various limitations. The study relies on cross-sectional data and is correlational, which makes it difficult to infer

causation. The possible problem of endogeneity, reverse causality and omitted variable bias are not clearly stated. Moreover, data quality in different countries can also be different, which can affect model results. Longitudinal data and causal inferences, including instrumental variables or panel models, should be included in future studies to more clearly establish directional relationships. Additional extensions can also examine how emerging technologies such as AI-blockchain synergy can be integrated and whether they can improve the financial inclusion of the underrepresented and rural population.

Author Contributions S.C. conceived the study, developed the FERENet framework, performed the data collection and analysis, and wrote the original draft of the manuscript. S.C. also prepared all the figures and tables. The author has read and approved the final version of the manuscript.

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Data Availability The dataset and reproducibility of code available at: [https://github.com/VisionLangAI/Digital-Literacy-and-its-Impact-Analysis](https:/github.com/VisionLangAI/Digital-Literacy-and-its-Impact-Analysis).

Declarations

Competing Interests The authors declare no competing interests.

Conflict of Interest The authors declare that they have no conflicts of interest.

Consent to Participate Not applicable.

Consent to Publish Not applicable.

Ethics Not applicable.

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